Jonathan Pan

j___@uci.edu 626 xxx xxxx linkedin.com/in/jonathanypan github.com/jonnypan2325 jonathanpan.me U.S. Citizen

Education

University of California, Irvine *Expected June 2027*
B.S. in Computer Science GPA: 3.81/4.0
- **Coursework:** Data Structures & Algorithms, AI/ML, Data Mining, Database Management, Computer Vision

Experience

MyCase (8am) *June – Sept 2026*
Software Engineering Intern *San Diego, CA*
- Developed Quickview, a document previewer in **Rails**, **React**, and **TypeScript**, cutting **p95 latency to 1.7 seconds**.
- Replaced **pdf.js** range fetching with a parallelized multipart **S3** download, reducing PDF fetches from **100 requests to 4**.
- Designed **Mixpanel telemetry** to identify preview bottlenecks, tracking latency, failure causes, and user abandonment.

Canon U.S.A. *Sept 2025 – June 2026*
Software Engineering Intern – Future Services Team *Irvine, CA*
- Delivered a **GCP** document platform using **Gemini on Vertex AI** to classify and extract data for **5 production tenants**.
- Shipped lease-based **Firestore** locks with real-time **WebSocket** updates to prevent conflicting document edits.
- Engineered the AI classification and extraction pipeline to a **<8% error rate** across **50+ document types**.
- Cut document cold-load latency from **30s to 14s** by deferring page rendering and parallelizing AI classification.
- Integrated the platform with Canon's enterprise document management system, shipping reliable large-file uploads.

Canon U.S.A. *June – Sept 2025*
Software Engineering Intern – Service Automation Tools *Irvine, CA*
- Developed an AI (**RAG**) support chatbot on **Azure** with **300K+** service manuals and device specific diagnostics.
- Cut AI inference costs by **40%** with document filtering, prompt compression, and an agent harness preventing loops.
- Tuned retrieval and prompts through **A/B tests**, validating changes against technician resolution outcomes.

U.C. Irvine – BIASLab *April 2024 – Present*
Student Researcher *Irvine, CA*
- Engineered a **PyTorch CycleGAN** for privacy-preserving object detection, retaining **87%** of original accuracy (mAP).
- Designed a spectral decoding framework for **audio language models**, reducing input token counts.
- Created and annotated a **2,000+** image dataset for evaluating privacy-preserving computer vision models.

Altek Medical *July – August 2024*
Software Engineering Intern *Hsinchu, Taiwan*
- Built a **Java** DICOM pipeline for 1080p video, reducing latency to **<200ms** per frame to support real-time data streaming.
- Implemented **TLS** encryption to securely archive patient media and metadata in a **PostgreSQL** PACS server.

Leadership

Projects Chair | *Artificial Intelligence at UCI* *July 2024 – Present*
- Authored an ML tutorial series (linear classifiers → **CNNs**) and mentored 50+ students on projects in **Python**/**PyTorch**.
- Taught members to build **multi-agent workflows** for coding with verification loops and multi-role decomposition.

Projects

Street-Art | *Python, Node.js, Tailwind, Firebase, Git*
- Created a web app transforming uploaded images into navigable GPS routes ("GPS Art") for Strava runners and cyclists.
- Implemented **Iterative Closest Point (ICP)** to align sketches with street topology and **NetworkX** for route optimization.
- Established automated benchmarks and human A/B reviews to assess route accuracy, performance, and visual quality.

Vroommates | *Python, React, Flask, Firestore, Google Cloud, Vercel, Git*
- Built a carpool platform that matches drivers and riders using route algorithms with Google Maps, **Flask**, and **React**.
- Persisted ride, rider, and match state in **Firestore**, supporting concurrent users across sessions and devices.
- Implemented **Google ID-token verification** and account-scoped APIs for saved routes and locations.

Skills

Languages: Python, TypeScript, JavaScript, Ruby, Java, C++, SQL
AI/ML: Vertex AI (Gemini), PyTorch, LangGraph, RAG/GraphRAG (Neo4j), OpenAI embeddings
Cloud/DBs: GCP (Cloud Run, Firestore, GCS), AWS (S3, Lambda), Azure, PostgreSQL, Redis, Terraform (IaC), Docker
Frameworks/Tools: Rails, React, FastAPI, Node.js, Next.js, Jest, Vitest, Playwright, GitHub Actions CI/CD